

1 Introduction

This report summarizes the progress achieved during the current reporting period for the UADE Racing Team BFMC vehicle. During this period, the main focus was to consolidate the transition from a more classical lane-keeping approach toward a local AI-based architecture centered on deep segmentation, physically interpretable lane reconstruction, and Stanley-based lateral control. The objective was not only to improve raw detection quality, but to build a more robust and traceable stack able to operate under embedded constraints, challenging lighting, partial lane loss, and demanding track geometries.

2 Planned activities

- Improve lane detection robustness under challenging lighting conditions and worn lane markings.
- Consolidate the transition from classical image-processing heuristics to AI-based local segmentation.
- Refine physical lane reconstruction to obtain metric crosstrack error and preserve continuity when one lane boundary is missing.
- Further tune the Stanley-based lateral controller for smoother and more stable tracking, especially in sharp turns.
- Integrate obstacle- and sign-aware behavior into the perception pipeline.
- Continue practical integration of additional sensors (LiDAR and ToF) on the physical vehicle.
- Validate the updated stack in controlled tests and real track runs.

3 Status of planned activities

Lane detection robustness and AI-local perception

Status: Ongoing.

Implementation / results: The perception stack is now centered on an AI Local pipeline running entirely on embedded hardware. Instead of relying on manual pre-processing stages such as HSV thresholding, edge detection, Hough transforms, or sliding-window heuristics, the system uses a TensorRT-optimized segmentation model that directly estimates the left and right lane masks, while also detecting relevant traffic signs in the same inference pass. This transition improved robustness under non-uniform illumination and simplified the perception architecture by consolidating it around a single primary path.

Difficulties / notes: Robustness improved significantly, but extreme glare, saturation, and domains poorly represented during training can still degrade performance.

Physical lane reconstruction and fallback when one boundary is missing

Status: Completed / under validation..

Implementation / results: The segmented lane masks are not used directly as control commands. Instead, they are converted into a physically interpretable lane geometry. The system reconstructs the lane center, estimates heading from geometric variation, converts pixel observations into metric quantities using the known BFMC lane width, and computes a direct physical crosstrack error. When only one lane line remains visible, the missing side is reconstructed from the nominal track width, allowing the vehicle to preserve continuity within the same geometric framework rather than switching to ad hoc heuristics.

Difficulties / notes: Very degraded one-line scenarios and extreme curves still require additional validation and tuning of safety margins.

Lateral control tuning

Status: Ongoing.

Implementation / results: Lateral control is now based on Stanley as the single operational controller, fed by physically meaningful lateral error instead of relying only on image-space deviation. The implemented formulation extends standard Stanley with practical stabilizing terms, including curvature compensation, steering damping, yaw-related correction, actuator saturation, and output post-processing. When the physical crosstrack estimate is reliable, the controller can prioritize that metric directly, reducing sensitivity to perspective-induced angular noise.

Difficulties / notes: The remaining challenge is to balance responsiveness and smoothness in aggressive curves and at higher operating speeds without over-filtering genuine corrections.

Adaptive parametrization and continuity under difficult conditions

Status: In progress.

Implementation / results: The project increasingly uses adaptive, rule-based parametrization rather than switching between different perception systems. The same AI Local + physical geometry + Stanley core is preserved, while parameters such as lookahead position, heading limits, safety margins, and temporal filtering aggressiveness can be adjusted according to the observed context. This makes the vehicle more consistent under lighting changes, lane loss, and curve severity while keeping the system explainable and traceable.

Difficulties / notes: Adaptive tuning is still being refined to avoid unnecessary conservatism or abrupt behavior changes.

Obstacle/sign integration and real-time validation

Status: In progress.

Implementation / results: Sign detections already share the same local perception pipeline and are transformed into validated behavior-level events. In parallel, the stack has been optimized for embedded execution through a TensorRT engine, last-only frame policy, and reduced latency across the critical path, supporting operation near 25 FPS in stable conditions.

Difficulties / notes: Broader on-track validation is still needed to fully characterize performance across all lighting and track conditions.

4 General status of the project

At the current stage, the vehicle already has an operational lane-keeping stack based on local AI segmentation, physical lane reconstruction, metric crosstrack estimation, one-line fallback, temporal filtering, and an extended Stanley controller. The main qualitative improvement is that the solution can no longer be described as a strengthened OpenCV-style lane detector, but as an integrated embedded architecture where perception, geometry, filtering, and control are designed together. This provides a more advanced and robust basis for BFMC operation, especially under variable lighting, reflections, partial occlusions, and tight curves.

The main remaining gaps are broader experimental validation, further refinement of adaptive parameters, and deeper integration of sensor-assisted maneuvers and behavior-level decision logic.

5 Upcoming activities

- Finalize validation of the AI Local stack under hard lighting and worn-lane conditions.
- Continue tuning the extended Stanley controller in sharp curves and higher-speed scenarios.
- Refine adaptive rule-based parametrization for context-dependent robustness.
- Deepen integration of sign/obstacle-aware behavior and sensor-assisted maneuvers.
- Run additional repeatable on-vehicle experiments with synchronized telemetry, replay, and comparative metrics.